

RANGANATHAN KARPAGAM

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WORK EXPERIENCE

Software Engineer II – Microsoft Corporation, Atlanta

June 2022 - Present

- Architected and shipped a security platform within Microsoft Intune (C#/TypeScript, Scala) processing billions of events/day at 99.99% uptime, powering the dashboard 1,000+ enterprise customers use daily to configure security access rules, manage device settings, and track device compliance progress.
- Redesigned the platform's data pipeline architecture — restructuring how usage events are processed and batched — cutting latency 87% (4min to <30sec) and unlocking 10x customer growth without added infrastructure cost.
- Re-architected client-side event aggregation — batching and pre-aggregating high-volume events before transmission instead of sending them individually — replacing an inefficient per-event pipeline and cutting downstream data volume 98%.
- Led expansion to multiple sovereign cloud regions, coordinating 5 engineering teams and building a repeatable deployment playbook now used for all regional rollouts — enabling 15+ regulated organizations to onboard.
- Modernized the admin console from legacy Knockout.js to React, using GitHub Copilot to accelerate development, meeting accessibility standards and driving a 30% increase in admin adoption.
- Spearheaded security compliance initiatives, eliminating manual secrets through managed identity migration and certificate-based authentication via Azure Key Vault — Freed 20 engineering hours per sprint and removed secret exposure risk.
- Built an AI-powered diagnostic tool that lets engineers query production health in plain language instead of manual log-digging— DB health queries, time-interval pipeline degradation detection, and contextual log interpretation; reduced investigation-to-root-cause time by 70% and enabled new engineers to independently diagnose production incidents without tribal knowledge.
- Introduced AI-assisted coding tools (Claude) into the team's workflow — restructured shared codebase conventions to work well with AI tooling and helped 3 other engineers ramp up on it, cutting routine PR turnaround time by roughly 20%.
- Eliminated UI blocking on long-running report generation by building an asynchronous Node.js/TypeScript processing service — keeping the dashboard fast and responsive for users.

Application Developer, Tata Consultancy Services – Humana Inc, Louisville

October 2019 - June 2022

- Developed real-time IT Service Management dashboards using AngularJS and D3.js, providing 200+ SREs daily visibility into service health — enabling them to identify performance bottlenecks 30 minutes earlier and respond faster to infrastructure issues.
- Built an automated monitoring system using Dynatrace to detect infrastructure issues like memory and CPU saturation, automatically creating ServiceNow incidents for the response team — reducing outage risk and improving incident response time.
- Owned Survey Portal, a company-wide compliance tool for employee onboarding, built with ASP.NET and Bootstrap — enabling systematic tracking and audit of new hire processes across the organization.

Software Engineer, Larsen & Toubro Infotech, India

June 2015 - July 2017

- Developed an online assessment tool adopted across all campus recruitment drives, supporting 1,000+ candidates annually — engineered a randomization algorithm to eliminate plagiarism risk and ensure fair, consistent evaluation across cohorts.

EDUCATION

M.S. Computer Science, Illinois Institute of Technology (2019)

B. Tech Information Technology, Anna University (2015)

TECHNICAL SKILLS

Languages: C#, Scala, JavaScript, SQL, Java

Frontend: React, NodeJS, AngularJS, HTML5, CSS, D3.js

Cloud: Azure, Managed Identities, Key Vault

Tools: ServiceNow, CI/CD, GitHub Copilot, Claude Code CLI, MCP, LLM API integration, prompt engineering